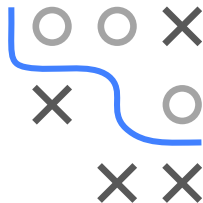


Introduction to Machine Learning

Gaussian Processes

Stochastic Processes and Distributions on Functions



$f(x)$



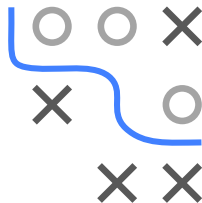
$\sim \mathcal{N}(\mathbf{m}, \mathbf{K})$

Learning goals

- GPs = distributions over functions
- Marginalization property
- Mean and covariance function

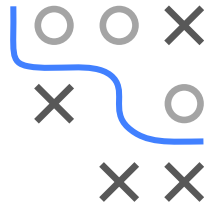
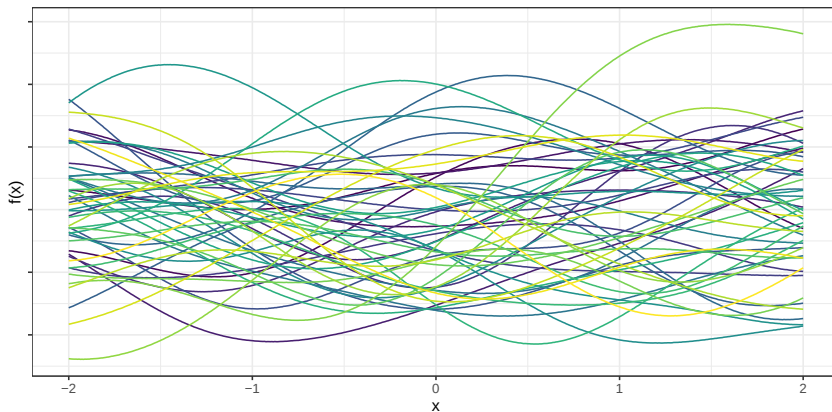
FUNCTION-SPACE VIEW

- New POV: rather than finding θ which parameterizes $f(\mathbf{x} \mid \theta)$, search in space of admissible functions directly
- Sticking to Bayesian inference, specify prior distribution **over functions** and update according to observed data points



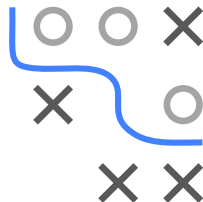
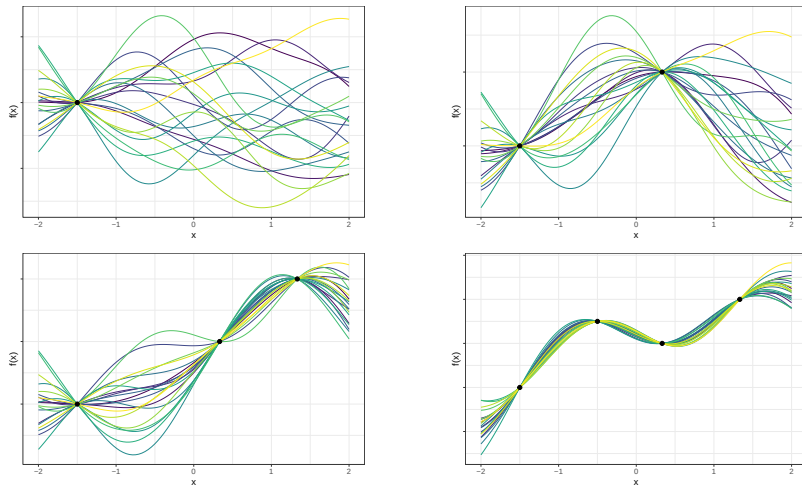
DRAWING FROM FUNCTION PRIORS

- Imagine we could draw functions from some prior distribution



DRAWING FROM FUNCTION PRIORS

- Restrict sampling to functions consistent with observed data



- Variety of admissible functions shrinks with seeing more data
- Intuitively: distributions over functions have “mean” & “variance”

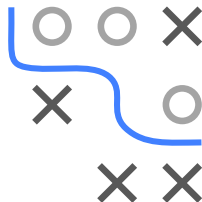
WEIGHT-SPACE VS. FUNCTION-SPACE VIEW

Weight-Space View

- Parameterize functions
(e.g., $f(\mathbf{x} \mid \theta) = \theta^\top \mathbf{x}$)
- Define distributions on θ
- Inference in param space Θ

Function-Space View

- Work on functions directly
- Define distributions on f
- Inference in fun space $\mathcal{H}$

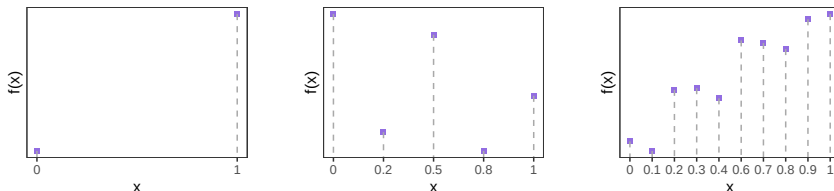


DISCRETE FUNCTIONS

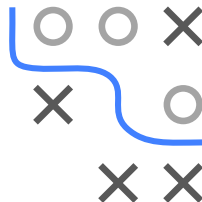
- Let $\mathcal{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$, $\mathcal{H} = \{f \mid f: \mathcal{X} \rightarrow \mathbb{R}\}$
- Any $f \in \mathcal{H}$ has finite domain with $n < \infty$ elements
 $\Rightarrow$ neat representation with n -dim vector

$$\mathbf{f} = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(n)})]^T$$

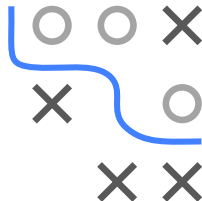
- Example functions living in this space for $|\mathcal{X}| \in \{2, 5, 10\}$



- NB: The $\mathbf{x}^{(i)}$ in the above are not really training points, we don't even consider training here. They are the points where we measure our (here: 1D) discrete functions. However, to avoid inventing too many symbols, and since the whole notation leads nicely into what follows next, we accept this "abuse" here.



DISTRIBUTIONS ON DISCRETE FUNCTIONS



- Specify density on vectors / functions with finite domain $f \in \mathcal{H}$
- Natural way: vector representation as n -dim RV, e.g.,

$$\mathbf{f} = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(n)})]^T \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

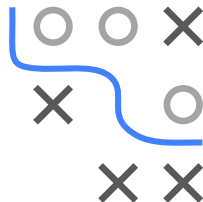
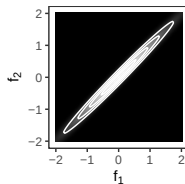
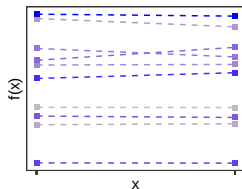
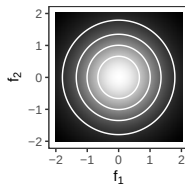
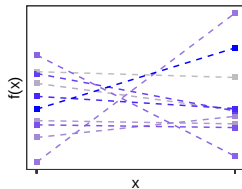
- For now: set $\mathbf{m} = \mathbf{0}$, assume $\mathbf{K}$ to be given

EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Example ctd: $\mathbf{f}$ on 2 points
- Sample representatives by sampling from a 2-dim Gaussian

$$\mathbf{f} = [f(\mathbf{x}^{(1)}), f(\mathbf{x}^{(2)})]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$$

- Where points are not (top) or strongly (bottom) correlated
- RHS shows 2D density

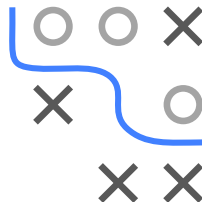
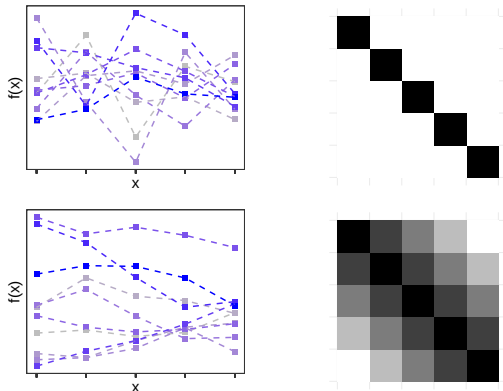


EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Example ctd: $\mathbf{f}$ on 5 points
- Sample representatives by sampling from a 5-dim Gaussian

$$\mathbf{f} = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(5)})]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$$

- Where points are not (top) or strongly (bottom) correlated
- RHS shows correlation matrix / structure

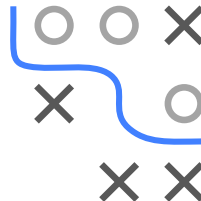
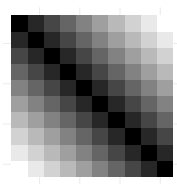
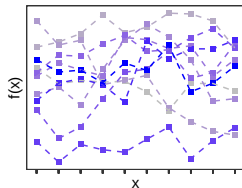
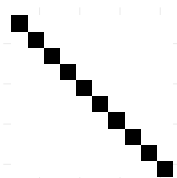
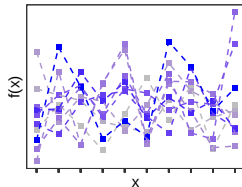


EXAMPLE: RANDOM DISCRETE FUNCTIONS

- Example ctd: $\mathbf{f}$ on 10 points
- Sample representatives by sampling from a 10-dim Gaussian

$$\mathbf{f} = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(10)})]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$$

- Where points are not (top) or strongly (bottom) correlated
- RHS shows correlation matrix / structure



SPATIAL CORRELATION

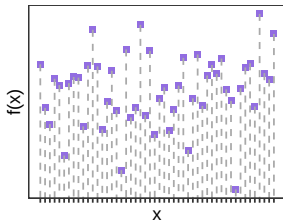
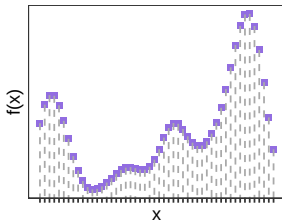
- “Meaningful” functions (on numeric $\mathcal{X}$) often have spatial property:

$\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$ close in $\mathcal{X} \Rightarrow f(\mathbf{x}^{(i)}), f(\mathbf{x}^{(j)})$ close / strongly correlated in $\mathcal{Y}$

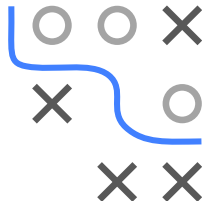
- In other words: fun. values of nearby points should be correlated
- Enforce this by choosing dist.-based covariance function

$\mathbf{x}^{(i)}, \mathbf{x}^{(j)}$ close in $\mathcal{X} \Leftrightarrow \mathbf{K}_{ij}$ high

- E.g., $\mathbf{K}_{ij} = k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) \left(-\frac{1}{2}\|\mathbf{x}^{(i)} - \mathbf{x}^{(j)}\|^2\right)$ vs identity cov.



- More on covariance function, or **kernel**, $k(\cdot, \cdot)$ later on

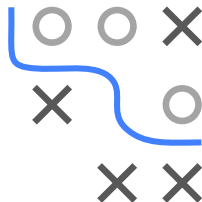


FROM DISCRETE TO CONTINUOUS FUNCTIONS

- So far: Multivar Gaussians to model outputs of discrete functions

$$\mathbf{f} = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(n)})]^T \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- Can we simply extend our distribution def to **continuous**-domain functions by taking $n \rightarrow \infty$?
- Unclear how to obtain “infinitely” long (Gaussian) random vectors
- Observation: random vectors $\mathbf{f}$ are collections of RVs enumerated by $\{1, \dots, n\} \Rightarrow$ **indexed family**
- Can we use more general, infinite index sets?



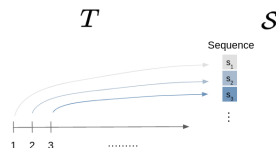
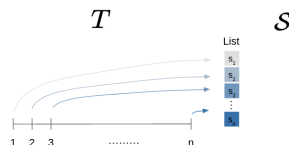
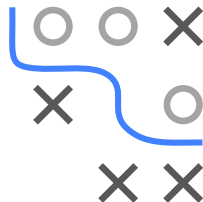
DEFINITION: INDEXED FAMILY

- Index T allows us to identify objects in arbitrary sets $\mathcal{S}$

$$s : T \rightarrow \mathcal{S}, \quad t \mapsto s_t = s(t)$$

- This mapping is the formal definition of notation $\{s_t : t \in T\}$
- Example: real-valued $\mathcal{S}$

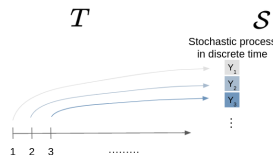
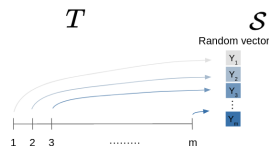
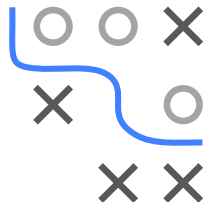
- $\mathcal{S} = \mathbb{R}, t \mapsto s_t$
- Finite index set, e.g.,
 $T = \{1, \dots, n\} \Rightarrow$ vector
- Countable, infinite index set,
e.g., $T = \mathbb{N} \Rightarrow$ sequence
- Uncountable index set, e.g.,
 $T = \mathbb{R} \Rightarrow$ function



DEFINITION: STOCHASTIC PROCESS

- Collection (potentially infinite) of RVs as indexed family $\{Y_t : t \in T\}$; further distributional assumptions give rise to important subclasses
- Intuition: probability distributions describe random vectors, SP describe random functions
- Examples

- $\mathcal{S}$: space of RVs, $t \mapsto Y_t$
- Finite index set, e.g.,
 $T = \{1, \dots, m\}$
 $\Rightarrow$ random vector
- Countable, infinite index set,
e.g., $T = \mathbb{N} \Rightarrow$ discrete-time SP
- Uncountable index set, e.g.,
 $T = \mathbb{R} \Rightarrow$ continuous-time SP



DEF.: GAUSSIAN PROCESS

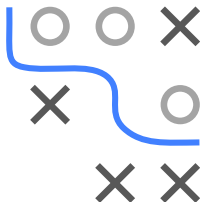
► Rasmussen and Williams 2006

► Snelson 2001

- Special kind of SP with index set $\mathcal{X}$; often $\mathcal{X} = \mathbb{R}^p$, but as in SVMs, feature vectors only enter the model via the kernel, so we can work on arbitrary spaces
- We write formally $f \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$
- Defining marginalization property: we have a GP iff for any finite set of inputs $\mathbf{X} \subset \mathcal{X}$,

$$f(\mathbf{X}) \sim \mathcal{N}(m(\mathbf{X}), k(\mathbf{X}, \mathbf{X}))$$

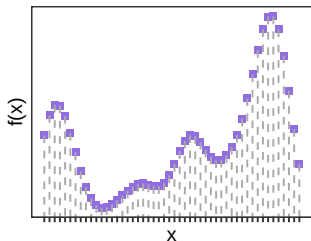
- With **mean function** $m : \mathcal{X} \rightarrow \mathbb{R}$ and **cov function** $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_0^+$
- With slight abuse of notation, we allow matrix args and write:
 - $\mathbf{m} = m(\mathbf{X}) = [m(\mathbf{x}^{(1)}), \dots, m(\mathbf{x}^{(n)})]^T$
 - $\mathbf{K} = k(\mathbf{X}, \mathbf{X}) = (k(\mathbf{x}, \tilde{\mathbf{x}}))_{\mathbf{x}, \tilde{\mathbf{x}} \in \mathbf{X}}$



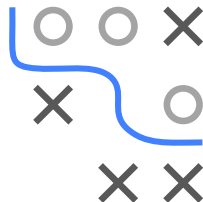
MARGINALIZATION PROPERTY

- For **any** finite set of inputs $\mathbf{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\} \subset \mathcal{X}$:

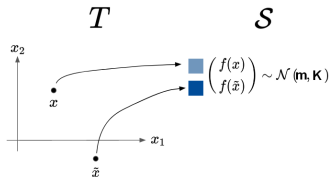
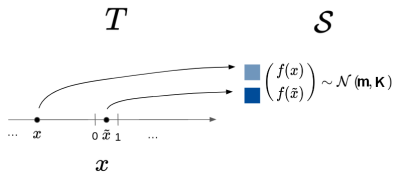
$$\mathbf{f} = f(\mathbf{X}) = [f(\mathbf{x}^{(1)}), \dots, f(\mathbf{x}^{(n)})]^T \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$



$$\begin{matrix} f(x) \\ \text{[gray square]} \\ \text{[gray square]} \\ \vdots \\ \text{[blue square]} \end{matrix} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$



- Example with 1D (left) and 2D (right) index set $\mathcal{X}$: Dimension of $\mathbf{f}$ depends on n , not on dimension of $\mathcal{X}$:



GP EXISTENCE THEOREM

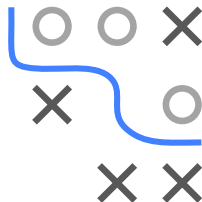
- For **any**
 - state space $\mathcal{X}$,
 - mean function $m : \mathcal{X} \rightarrow \mathbb{R}$,
 - covariance function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}_0^+$,

there **exists** $f \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$ s.t. $\forall \mathbf{x}, \tilde{\mathbf{x}} \in \mathcal{X}$

$$\begin{aligned}\mathbb{E}(f(\mathbf{x})) &= m(\mathbf{x}) \\ \text{Cov}(f(\mathbf{x}), f(\tilde{\mathbf{x}})) &= k(\mathbf{x}, \tilde{\mathbf{x}})\end{aligned}$$

and $f(\mathbf{X}) \sim \mathcal{N}(m(\mathbf{X}), k(\mathbf{X}, \mathbf{X}))$ for any $\mathbf{X} \subset \mathcal{X}$

- Version of Kolmogorov consistency theorem
 $\Rightarrow$ proof ▶ Grimmett and Stirzaker 2001 (Thm. 8.6.3)



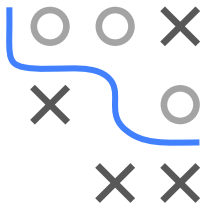
IMPLICATIONS OF EXISTENCE THEOREM

- GPs completely specified by their mean & cov function

$$m(\mathbf{x}) = \mathbb{E}[f(\mathbf{x})]$$

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \text{Cov}(f(\mathbf{x}), f(\tilde{\mathbf{x}})) = \mathbb{E}[(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})])(f(\tilde{\mathbf{x}}) - \mathbb{E}[f(\tilde{\mathbf{x}})])]$$

- For now, we consider zero-mean GPs with $m(\mathbf{x}) \equiv \mathbf{0}$
 $\Rightarrow$ common, not necessarily drastic assumption
- Denote by $\mathcal{GP}(\mathbf{0}, k(\cdot, \cdot)) \Rightarrow$ properties mainly governed by $k(\cdot, \cdot)$
- By virtue of existence thm: sampling from GP priors gives us random functions with our properties of choice

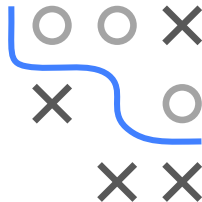
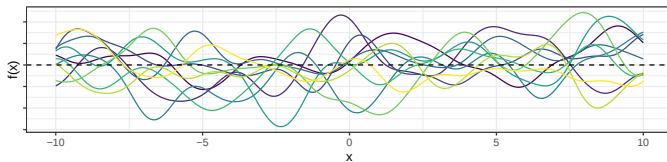


SAMPLING FROM GAUSSIAN PROCESS PRIORS

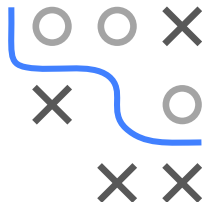
- Example: $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ with cov function

$$k(\mathbf{x}, \tilde{\mathbf{x}}) = \exp\left(-\frac{1}{2}\|\mathbf{x} - \tilde{\mathbf{x}}\|^2\right)$$

- To visualize sample functions,
 - choose high number n of points $\mathbf{X} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$
 - compute $\mathbf{K} = k(\mathbf{X}, \mathbf{X})$ from all pairs $\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \in \mathbf{X}$
 - draw $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$
- 10 randomly drawn functions (note 0 mean)



FURTHER READING



- Will go through many details now, but some general refs already
- The standard book: [▶ Rasmussen and Williams 2006](#)
- Good videos can be found here: [▶ Monk 2011](#) [▶ Freitas 2020](#)